

Goal: Build a compact automotive RF sensing system capable of detecting and classifying microwave radar emissions.

Phase 1

Target outputs:

Detected/not detected
Approximate frequency
Signal strength
Confidence level
Alert through display/buzzer
Logged measurements for later analysis

Engineering constraints:

<\$300 prototype
Fits inside a car
12-V automotive power
Real-time processing
Custom-designed enclosure
Semester completion

Phase 2

Learn RF Design

Observe RF signals with RTL-SDR or other sensing equipment with spectrum analysis

Write code to extract variables

frequency
signal power
bandwidth
duration
peak frequency

Phase 3

Build detector algorithm

Signal detected

Frequency: X GHz

Bandwidth: XX MHz

SNR: XX dB

Duration: XX ms

Classification: Candidate radar

Confidence: XX%

Phase 4

Design automotive package:

enclosure

PCB mounts

antenna mounts

windshield/dashboard mount

heat-management features

cable routing

vibration isolation